

Renewsys MES AI Chatbot — Phase 1 Software Requirements Specification (Final Refined)

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1. Document Control

Field	Value
Document Title	Renewsys MES AI Chatbot — Phase 1 Software Requirements Specification (Final Refined)
Project	Renewsys Chatbot
Customer	Renewsys India Pvt Ltd
Primary Plant (Phase 1)	Khopoli
Source Data	MES AI Chatbot — Requirements Responses.xlsx (139 fields; submitted 2026-09-25 21:17)
Stakeholder-Named SRS/UAT Approver	Developer (per V3 — preserved verbatim; organizational approval workflow should still be confirmed)
Document Version	3.0 (Final Refined)
Date	2026-09-26
Prepared For	Renewsys stakeholders, IT/DBA, Jinchen MES vendor/developer, implementation team

Revision History

Version	Date	Description
1.0	2026-09-26	Initial Phase 1 SRS draft
2.0	2026-09-26	First refinement; added full traceability matrix and stricter classification
3.0	2026-09-26	Second refinement: corrected Email/Microsoft Teams scope, source-of-truth wording, blank-field handling, and restructured document per finalized 40-

Version	Date	Description
		section outline

2. Requirements Baseline Status

Requirements Baseline Status: Partially confirmed; some items are assumptions.

This statement reflects the respondent's own self-assessment (questionnaire field FNL1: "*Partially confirmed; some items are assumptions*") and governs how this entire document must be read. This SRS is **not** a fully finalized requirements baseline.

Several technical and business definitions used throughout this document require validation with:

- **IT** — network, infrastructure, deployment, and operational questions.
- **DBA / MES technical owner** — database engine, schema, stored procedures, performance characteristics.
- **Jinchen MES vendor/developer** — the original MES build team, who may hold information not available to local Renewsys staff (existing integration architecture, APIs, connection mechanisms).
- **Relevant business/HOD stakeholders** — KPI formulas, business semantics (Material, quantity semantics, process loss, rework), and measurable business outcomes.

Every requirement, architectural note, and traceability entry in this document is classified using exactly one of the following labels, applied consistently and never upgraded to CONFIRMED merely because a value is technically reasonable:

- **CONFIRMED**
- **STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION**
- **AMBIGUOUS**
- **TBD**
- **ASSUMPTION**
- **PROPOSED TECHNICAL SOLUTION**
- **FUTURE PHASE**
- **OUT OF SCOPE**

Classification counts across all 139 questionnaire fields

Classification	Count
CONFIRMED	74
AMBIGUOUS	39
TBD	15
STAKEHOLDER-PROVIDED —	8

Classification	Count
REQUIRES TECHNICAL VERIFICATION	
FUTURE PHASE	1
N/A — Administrative	2

(ASSUMPTION and PROPOSED TECHNICAL SOLUTION items are not questionnaire responses themselves — they appear throughout this SRS wherever the implementation team fills a gap and are consolidated in Sections 32–34.)

An unanswered or content-free questionnaire field (e.g., a blank cell, or a response of “None” that provides no usable information) is **never** converted into “No,” “Not required,” or “Out of Scope” in this document. It is marked **TBD — No response provided**, and the underlying decision is left open until confirmed.

3. Executive Summary

Renewsys India Pvt. Ltd. wants a conversational assistant (the “Renewsys Chatbot”) that lets MES users at the Khopoli plant ask questions in plain English and receive information currently obtained by manually querying or reporting from the Jinchen MES database. Phase 1 targets a single plant, approximately 15–20 total users, and 3–4 concurrent users.

The stakeholder intent, in the respondent’s own words (A4): *“Instead of retrieving data directly from the database we can take help from this chat what it can translate from plain English to the queries and generate the required fields.”* This is a **stated business need**, not a technical design; how it is technically achieved (an LLM, a semantic layer, a Text-to-SQL pipeline) is a matter for the implementation team to propose, and any such proposal is explicitly labeled **PROPOSED TECHNICAL SOLUTION** throughout this document.

The respondents are MES **users**, not the developers of the underlying MES — which was built by a third-party (Jinchen) team whose server the respondents describe as being “in China” and something “we do not handle.” As a result, the database engine, schema, stored procedures, connection architecture, several business-data definitions, KPI formulas, and portions of the security/network environment are **not currently known** and are documented here as TBD rather than assumed.

This third revision of the SRS makes several corrections identified during review:

1. **Email and Microsoft Teams remain identified as in-scope systems** (Q1), not reclassified as Future Phase — their integration detail is TBD, which is a different status from being out of scope.
2. **The complete 139-field questionnaire traceability matrix** replaces any earlier “representative sample” table (Section 38).
3. **The Source-of-Truth register** no longer uses the word “presumed” — Jinchen MES is documented as a *candidate, not yet validated* source for each domain.

4. **Genuinely unanswered questionnaire fields are marked TBD**, not silently converted into “No”/“None”/“Out of Scope.”
 5. **The ambiguity between the MES server being “in China”/unmanaged locally and the application being planned on the company LAN with direct connectivity** is documented as an open technical issue rather than silently reconciled.
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4. Problem Statement

Stakeholder-provided (A4, CONFIRMED as a statement of intent): *“Instead of retrieving data directly from the database we can take help from this chat what it can translate from plain English to the queries and generate the required fields.”*

Users currently retrieve MES information manually. No quantified pain point (cost of delay, error rate, hours lost) was supplied, and none is invented here.

5. Objectives and Scope

5.1 Business Objective

Stakeholder intent (derived from A4): enable users to ask MES-related questions in natural language and retrieve relevant information without directly querying the database.

Classification: STAKEHOLDER-PROVIDED / DERIVED — measurable business outcomes TBD.

A5 (“What business outcomes are expected?”) was answered **“All”** — classified **AMBIGUOUS**. No quantified time savings, productivity improvement, cost reduction, or adoption target is invented in this SRS. Detailed, measurable success criteria require confirmation with the business stakeholders.

5.2 Phase 1 Scope Summary

See Section 7 for the full scope statement. In brief: single plant (Khopoli), read-only MES querying via natural language, session-scoped chat, the confirmed KPI/response-type/date-expression lists, and RBAC — all subject to the classifications recorded throughout this document.

6. Stakeholders and Users

Role	Source	Classification
Plant Manager	A6	CONFIRMED

Role	Source	Classification
Production Manager	A6	CONFIRMED
Quality Manager	A6	CONFIRMED
Production Planning	A6	CONFIRMED
Management / Director	A6	CONFIRMED
IT / System Administrator	A6	CONFIRMED

- Approximate total users: **~15–20** (A7, CONFIRMED as an approximate figure).
- Primary contacts: **“Managers, HOD”** (A9) — role labels only, no individual names or specific responsibilities. **Classification: AMBIGUOUS.** No names are invented; contact detail is TBD.

7. Phase 1 Scope

Confirmed Phase 1 scope elements:

- Single plant: **Khopoli** (A3, A8 — CONFIRMED).
- ~15–20 total users, ~3–4 concurrent users (A7, R1 — CONFIRMED as approximate).
- Natural-language question answering against MES data, **read-only** (F1, F2 — CONFIRMED).
- Session-scoped conversational memory (K5 — CONFIRMED).
- Natural-language date/time and shift expressions (L3, L4 — CONFIRMED), up to 30 days of history (L2 — CONFIRMED), ~24-hour data freshness (L1 — CONFIRMED).
- Response types: natural language, tables, KPI cards, charts, drill-down, and SQL visibility for Admin/IT only (I1, I3 — CONFIRMED).
- Export capability in general, exact formats TBD (I2 — AMBIGUOUS on format).
- Local application authentication with RBAC (G5, G6 — CONFIRMED); detailed role-to-data permissions TBD (G7, J2 — AMBIGUOUS).
- The KPI name list in Section 14 (M1 — CONFIRMED as names; formulas/source/targets NOT confirmed).
- MES, Email, and Microsoft Teams as systems in scope, per Section 8 (Q1 — CONFIRMED as in-scope systems; implementation detail TBD).

8. Systems in Scope

Confirmed (Q1): “MES, Email, Microsoft Teams.”

System	In Scope?	Integration Detail	Classification
MES (Jinchen MES)	Yes	Read-only query access;	CONFIRMED (scope) / TBD

System	In Scope?	Integration Detail	Classification
Email	Yes	schema/engine TBD No APIs, authentication, or workflow supplied (Q2 blank)	(technical detail) CONFIRMED (scope) / TBD (implementation)
Microsoft Teams	Yes	No APIs, authentication, or workflow supplied (Q2 blank)	CONFIRMED (scope) / TBD (implementation)

Email and Microsoft Teams **must not** be presented as confirmed Future Phase items — see Section 22 for the dedicated treatment of this correction.

9. Functional Requirements

9.1 Central Phase 1 Objective

Per V1 (CONFIRMED): “*Chatbot should answer the questions and create required chart/graphs etc. by reading the DB.*”

ID	Requirement	Priority	Source	Classification
FR-CORE-001	The system shall let users ask MES-related questions in natural English and receive an answer, table, KPI card, or chart derived from a read-only MES query.	High	V1, A4, F1, F2	CONFIRMED
FR-CORE-002	The system shall restrict database operations to SELECT only, using a dedicated read-only database account.	High	F2, F6	CONFIRMED
FR-CORE-003	The system shall support session-	Medium	K5	CONFIRMED

ID	Requirement	Priority	Source	Classification
	scoped conversational memory so follow-up questions can use prior context.			
FR-CORE-004	The system shall ask a clarifying question rather than guess when a question is ambiguous.	High	U1	CONFIRMED
FR-CORE-005	The system shall not always refuse to answer when required data is unavailable; the exact fallback behavior beyond this negative constraint is proposed.	Medium	U2	CONFIRMED (negative constraint) / PROPOSED (fallback design)
FR-CORE-006	The system shall restrict visibility of generated SQL/source query detail to Admin/IT users.	High	I3	CONFIRMED
FR-CORE-007	The system shall display data freshness / last-updated information with answers.	Medium	I4	CONFIRMED
FR-CORE-008	The system shall limit historical querying to the most recent 30 days unless otherwise approved.	Medium	L2	CONFIRMED
FR-CORE-009	The system shall	Medium	I2	CONFIRMED

ID	Requirement	Priority	Source	Classification
	provide export of results in general; exact formats are a proposed technical solution pending approval.			(capability) / PROPOSED (formats)
FR-CORE-010	The system shall distinguish, in its output, directly measured MES data from calculated/derived values, analytical interpretations, and recommendations/predictions.	High	N3, N4	CONFIRMED

How these are technically achieved is addressed in Section 23 (Architecture) as a Proposed Technical Solution, not itself part of the stakeholder requirement.

10. Natural-Language Query Processing

10.1 Confirmed Example Questions (E1)

Preserved, grammar lightly normalized, intent unchanged:

- “What was today’s production?”
- “How many units entered Glass?”
- “How many units are in Packing?”
- “From which line does this module get produced?”
- “At what time was the module produced?”
- “How many modules were created on this day?”
- “How many modules are produced on this line?”
- “How many scrap/bad units were found?”
- “How much production happened on this day?”

10.2 Broad Question Categories (STAKEHOLDER-PROVIDED — AMBIGUOUS)

The stakeholder answered “**All**” for the following categories, which is preserved as a high-level, non-exhaustive scope statement rather than expanded into invented detailed requirements:

- Quality/defect questions (E2)
- WIP/process-movement questions (E3)
- Shift/line/area comparison questions (E4)
- Date-range/historical questions (E5)
- Management/executive questions (E6)

Detailed question patterns and UAT acceptance criteria for each category **require clarification** with the stakeholders.

11. Query Interpretation and Clarification

Confirmed (U1): *Ask a clarification question rather than guess.* The system shall:

1. Detect ambiguity in the incoming question.
2. Ask a clarifying question rather than proceed on a guess.
3. Avoid generating a potentially misleading query while the interpretation is unresolved.
4. Execute the query only once the interpretation is sufficiently clear.

Examples of ambiguity requiring clarification (Proposed, illustrative): ambiguous date range, ambiguous line, ambiguous shift, ambiguous product/material, ambiguous quantity definition, ambiguous KPI reference.

Must-not-answer criteria (E7) and clarification-trigger criteria beyond U1 (E8) were both answered “**Over to you**” — classified **AMBIGUOUS**, i.e., delegated to the implementation team. Any resulting policy is a **PROPOSED** safe-answering approach, not a stakeholder-confirmed rule: avoid answering when the requested information is outside authorized data, the required field is unavailable, the query remains ambiguous after clarification, confidence is insufficient, the data source cannot support the answer, or the requested operation is not permitted (e.g., a write).

The system must never invent an answer when underlying business definitions are unknown (e.g., Material, process loss, rework — see Section 20).

12. Database Access

12.1 Confirmed Stakeholder Requirements

- Phase 1 is **read-only now**; transactional actions “may be added later” — not a Phase 1 commitment (F1, CONFIRMED).

- **SELECT-only** operations (F2, CONFIRMED).
- **No database writes** required (F3, CONFIRMED).
- Stakeholder reports required tables/views as **“Fully accessible”** (F4, CONFIRMED as stated) — see caveat below.
- No specific tables/fields identified as never-accessible (F5: “No” — CONFIRMED, no restriction named).
- **Dedicated read-only database account required** (F6, CONFIRMED).

12.2 Important Caveat on “Fully Accessible”

“Fully accessible” (F4) does **not** remove the need for least-privilege validation, credential security, a database permissions review, and production security approval before go-live. It reflects the absence of a *named* restriction from this respondent, not an instruction to bypass standard access governance. No table-level restriction is invented; any governance layer applied on top of F4 is a **PROPOSED TECHNICAL SOLUTION**.

12.3 Database Technology

B1 (“Database technology”) = “Over to you.” Classification: AMBIGUOUS.

No database engine (SQL Server, MySQL, PostgreSQL, Oracle, MongoDB, or otherwise) is claimed or inferred — including from the port number (8000) supplied in B3, which is not diagnostic of engine type and is not used to infer one here.

Stated design response:

Database technology/engine is not confirmed. The application architecture shall use an adapter-based database access layer so the final implementation can be selected after IT/DBA/MES vendor confirmation.

This is a **PROPOSED TECHNICAL SOLUTION**.

12.4 Server Location and Network Path — Open Technical Issue

The questionnaire contains a statement that has not been reconciled and is documented as an explicit open issue rather than silently resolved:

B6 states: *“Mes server is in China we do not handle”* — i.e., the MES server is described as being in China and not managed by the respondent. **G1/G2 state:** the planned FastAPI/application server location is *“same company LAN / server room”* with *“Direct LAN”* connectivity to the database.

This SRS does not assume what “China” means in this context — it is not assumed to indicate a purely local server, a cloud-hosted server, a remote vendor-hosted server, or any other specific topology. The physical/network location and ownership/access path of the Jinchen MES server require clarification. IT/MES vendor confirmation is required to determine whether **10.69.12.20** is directly reachable from the intended application server on the company LAN, and what network/security path (VPN, dedicated link, firewall rules, etc.) is actually required to reach it. This is carried forward as Open Issue OI-04 (Section 35).

12.5 Database Discovery — Proposed, Not Stakeholder-Named

Because the schema must be requested from IT/DBA (C1, CONFIRMED) and the engine is undetermined (B1, AMBIGUOUS), a controlled, read-only schema-discovery mechanism may be used by the implementation team. This is a **PROPOSED TECHNICAL SOLUTION**, not a capability named by the stakeholder. It must never write, alter, or delete data, and must never execute arbitrary user- or LLM-generated SQL against the production MES.

13. SQL Generation and Safety

13.1 Confirmed Controls

Control	Source	Classification
SELECT-only database operations	F2, U4	CONFIRMED
Dedicated read-only database account	F6, U4	CONFIRMED

13.2 SQL Validation — Explicitly Not Requested

U3 (“Should generated SQL be validated before execution?”) = “No.” This SRS does **not** state that Renewsys explicitly requested a SQL validation/parser layer. Any statement to the contrary in an earlier draft is corrected here.

13.3 Proposed Technical Safety Controls

The following are engineering recommendations proposed to safely implement the confirmed read-only requirement — none are stakeholder-requested, and each must be labeled as such if adopted:

PROPOSED TECHNICAL SAFETY CONTROL - SQL parser/validator - Statement-type enforcement (blocking DDL/DML at the application layer) - Query timeout - Resource/result-size limits - Database-side permission enforcement (in addition to the confirmed read-only account)

Never permitted regardless of any proposed control: INSERT, UPDATE, DELETE, DROP, ALTER, TRUNCATE, CREATE, GRANT, REVOKE, or administrative commands (consistent with F1/F2/F3).

14. KPI Requirements

14.1 Confirmed KPI Names (M1)

Total production, Get-in quantity, Departure quantity, Bad quantity, Scrap quantity, Yield, Defect rate, Scrap rate, Process loss, OEE, Availability, Performance, Quality, Downtime, Cycle time, WIP, Throughput, First Pass Yield, Rework rate.

KPI names are stakeholder-requested. Official calculation formulas, numerator/denominator definitions, exclusions, aggregation rules, and business ownership are **not** supplied in the questionnaire and must be validated before production use.

14.2 KPI Validation Table

KPI	Proposed Formula	Required Source Fields	Confirmed by Renewsys?	Validation Owner
Total Production	PROPOSED — sum of validated production/get-in events	Production/Get-in event count	No	Business/MES SME
Get-in Quantity	PROPOSED — count of Get-in events per D4 definition	Get-in station events	No	Business/MES SME
Departure Quantity	PROPOSED — count of Departure events per D5 definition	Departure station events	No	Business/MES SME
Bad Quantity	PROPOSED — count of Bad-flagged units per D6 definition	Quality/defect events	No	Quality Manager
Scrap Quantity	PROPOSED — count of Scrap-flagged units per D7 definition	Scrap events	No	Quality Manager
Yield	PROPOSED — $\text{Good units} \div \text{Total units}$	Production + Quality data	No	Business/MES SME
Defect Rate	PROPOSED — $\text{Bad units} \div \text{Total units}$	Quality data	No	Quality Manager
Scrap Rate	PROPOSED — $\text{Scrap units} \div \text{Total units}$	Scrap data	No	Quality Manager

KPI	Proposed Formula	Required Source Fields	Confirmed by Renewsys?	Validation Owner
Process Loss	Not proposed — depends on unresolved D11	Undetermined	No	Business/MES SME
OEE	PROPOSED — Availability × Performance × Quality	Downtime, throughput, quality data	No	Production Manager
Availability	PROPOSED — Run time ÷ Planned production time	Downtime data	No	Production Manager
Performance	PROPOSED — (Ideal cycle time × count) ÷ Run time	Cycle time + counts	No	Production Manager
Quality (OEE component)	PROPOSED — Good units ÷ Total units	Quality data	No	Quality Manager
Downtime	Not proposed — source unconfirmed	Downtime event log (existence unconfirmed)	No	Production Manager
Cycle Time	Not proposed — source unconfirmed	Process timestamps	No	Production Manager
WIP	Not proposed — depends on unresolved D10	Undetermined	No	Business/MES SME
Throughput	Not proposed — depends on confirmed production definition	Production events	No	Production Manager
First Pass Yield	Not proposed — depends on unresolved D12	Undetermined	No	Business/MES SME
Rework Rate	Not proposed — depends on unresolved D12	Undetermined	No	Business/MES SME

Every “Proposed” formula in this table is labeled **PROPOSED — REQUIRES BUSINESS VALIDATION** and must never be presented elsewhere in this document, or to any reviewer, as a Renewsys-approved formula.

14.3 KPI Source of Truth

M3 = “Over to you.” Classification: AMBIGUOUS.

KPI source-of-truth ownership is **TBD**. The final source and calculation ownership shall be established with the business/MES/IT stakeholders.

No source-of-truth system is invented (see also Section 21).

14.4 KPI Thresholds and Targets

M4 = “Over to you.” Classification: AMBIGUOUS.

No target value, or green/yellow/red threshold band, is invented. If the architecture proposes a configurable-thresholds mechanism, it is labeled **PROPOSED TECHNICAL SOLUTION**; actual target values must be supplied and approved by the appropriate business owner.

15. Analytics

Confirmed capabilities (N1): day-over-day comparison, shift comparison, line comparison, process comparison, product/material comparison, trend analysis, top loss processes, top defect processes, anomaly detection, root-cause investigation support, and what-if calculations.

Measured vs. calculated distinction (N3, CONFIRMED — “Yes”): output must clearly separate source data (directly retrieved), derived/calculated data, analytical interpretation, and recommendation/prediction.

Recommendations/predictions (N4, CONFIRMED — “Yes, informational only”): may be included but must never be represented as automated commands, guaranteed predictions, operational instructions, or transactional actions.

Advanced predictive analytics: V7 (future features) is blank. Advanced predictive capabilities are **not** introduced as a confirmed Future Phase item on that basis. They are classified **PROPOSED / TBD**, and detailed predictive-model requirements require later validation if and when the business chooses to pursue them.

Root-cause investigation support is interpreted as *“support through correlated operational data and drill-down analysis,”* not a claim of autonomous causal inference; several underlying inputs (process loss, rework) remain unresolved (Section 20) and limit what can currently be delivered.

16. Charts and Visualization

Required response types (I1, CONFIRMED): natural-language answer, tables, KPI cards, bar charts, line charts, pie/donut charts, trend charts, drill-down details, and source/query details for Admin/IT users only (I3, CONFIRMED).

Export (I2) = “All.” Classification: **AMBIGUOUS** on exact format. Proposed formats (e.g., CSV/Excel/PDF) are not confirmed; any implementation choice must be labeled **PROPOSED**, **pending implementation approval**.

Branding/UI (I5) = “Yes.” Confirmed as intent; no specific branding guidelines were supplied — exact visual specification is **TBD**.

Freshness (I4) = “Yes” — CONFIRMED; freshness/last-updated timestamp must accompany answers.

17. Conversational Context

Confirmed (K5): the assistant should remember previous messages within an active chat session (e.g., “What about KM2?” after a prior production question resolves using the earlier context). Interpreted, absent further specification, as **conversation context retained during an active session only** — persistent cross-session/long-term memory is not assumed or required.

18. Authentication and RBAC

Item	Response	Classification
Chatbot authentication mechanism	Local application accounts (G5)	CONFIRMED
RBAC required	Yes (G6)	CONFIRMED
Platform roles	Admin and Users (J1)	CONFIRMED
User’s plant/area/line/role known from login	Yes (J3)	CONFIRMED (requirement); mechanism TBD
Row-level security	No (J4)	CONFIRMED — not a Phase 1 requirement
Per-role data-domain permissions	“All” (J2) / “On the mes developer not on us” (G7)	AMBIGUOUS

J3 wording, precisely stated:

The system shall make the authenticated user's plant/area/line/role context available to authorization and query-processing components. Exact enforcement rules and permission mappings require confirmation.

This SRS does **not** claim that every query will automatically be filtered by plant/line unless and until that filtering logic is confirmed with the business/MES developer side. Detailed role-to-data-domain permissions remain owned by "the MES developer" per G7 and are not currently documented; Phase 1 therefore implements RBAC as a **configurable framework** (PROPOSED TECHNICAL SOLUTION), not hard-coded final permissions.

19. Security

Confirmed security-relevant items:

Item	Response	Classification
Public internet exposure of the database	Not permitted (G4)	CONFIRMED
External cloud AI processing of manufacturing data	Not permitted; data must stay in-network (K2)	CONFIRMED
Read-only database account	Required (F6)	CONFIRMED
SELECT-only operation	Required (F2)	CONFIRMED
RBAC	Required (G6)	CONFIRMED
Row-level security	Not required (J4)	CONFIRMED
Sensitive-value masking in logs	Required (T3)	CONFIRMED
Applicable compliance/security standards	No response provided (T4)	TBD — not interpreted as "no standards apply"

K2 stated formally:

Production AI processing shall not send MES/company data to an external cloud AI service unless the requirement is explicitly changed and approved.

No specific compliance/security standard (ISO 27001, SOC 2, GDPR, India's DPDP Act, or otherwise) is claimed to apply; T4 is a genuinely blank field and is marked **TBD**, not "no compliance standards."

20. Data Freshness and Time Handling

- Data freshness: **within approximately 24 hours** (L1, CONFIRMED). No real-time/streaming freshness is claimed.
- Maximum historical query scope: **30 days** (L2, CONFIRMED) — a stated chatbot *query-scope* limit; whether older MES data physically exists but is simply excluded from chatbot querying is **TBD**, not clarified by the stakeholder.
- Natural-language date questions required (L3, CONFIRMED); supported expressions (L4, CONFIRMED): Today, Yesterday, Current shift, Previous shift, Last 7 days, Last 30 days, This week, Last month, Custom date range, Custom time range.
- Shift timing (L6, CONFIRMED): Morning 07:00–15:00, Second 15:00–23:00, Night 23:00–07:00.
- Business definitions (D9, CONFIRMED): Shift = A/B/C; Line = KM1/KM2/KM3. ShiftName mapping (D1, CONFIRMED): A = Morning, B = Second, C = Night.
- **Timezone: L5 = “We don’t know.” Classification: AMBIGUOUS/TBD — IT/business confirmation required.** Asia/Kolkata (IST) is **not** assumed merely because the plant is located in India.

Other business definitions (preserved verbatim, D9/D4–D8)

Term	Definition
Area	Location
Workorder	Order which we get to produce
Start Time	Date/time of the report/process
End Time	Date/time “required” for the report/process
Getin	Module entering that station
Departures	Module leaving station
Bad	Module with defect
Scrap	Damaged/scrapped module
ExcludeDuplicates	Existing report behavior related to duplicate barcode scanner scans

Fields where the stakeholder said “IDK” (TBD — Business/MES Validation Required; no meaning invented)

- **Material** (D9)
- **Quantity semantics** — cumulative vs. transactional (D10)
- **Process loss** calculation method (D11)
- **Rework loops** — returns to earlier processes (D12)
- **Duplicate/repeated event semantics** beyond the confirmed barcode-scan issue (D13)

21. Integration Scope

This section formalizes the **Source-of-Truth Register**, replacing any earlier “presumed” language.

Q3 (“Which system is the source of truth for production, quality, inventory, work orders, and other domains?”) = “Not knowing.” Classification: AMBIGUOUS.

Source-of-Truth Register

Domain	Candidate Source	Confirmed?	Status
Production	Jinchen MES — not yet validated	No	TBD
Quality	Jinchen MES — not yet validated	No	TBD
WIP	Jinchen MES — not yet validated	No	TBD
Work Orders	Jinchen MES — not yet validated	No	TBD
Process events	Jinchen MES — not yet validated	No	TBD
Financial / Profit	TBD	No	TBD
Email/Teams integration data	TBD	No	TBD

Jinchen MES is documented as a **candidate source, not a confirmed source of truth**, for any domain. This is a deliberate correction from language such as “MES — presumed,” which overstates the current state of knowledge.

Profit specifically: MES availability of revenue, cost, margin, or profit information has **not** been established. See Section 30 (UAT) for the corresponding acceptance-criteria treatment.

22. Email and Microsoft Teams

This is a major correction from an earlier draft, which must not recur.

The questionnaire (Q1, CONFIRMED) explicitly lists Email and Microsoft Teams as systems in scope, alongside MES. **Email and Microsoft Teams must not be presented as automatically Future Phase**, and must not appear under a confirmed “Future Phase” list.

Item	Status
Identified as in-scope systems	CONFIRMED (Q1)
APIs/interfaces	Not supplied (Q2 blank) — TBD
Authentication mechanism	Not supplied — TBD

Item	Status
Authorization model	Not supplied — TBD
Message/report delivery workflow	Not supplied — TBD
Email-sending mechanism	Not supplied — TBD
Teams integration mechanism	Not supplied — TBD
Read-only vs. outbound integration	Not supplied — TBD
Exact Phase 1 implementation boundary	Not supplied — TBD
Technical feasibility and ownership	Not supplied — TBD

Correct statement of status:

Email and Microsoft Teams are identified by the stakeholder as systems in scope. However, the questionnaire does not provide APIs, authentication mechanisms, integration workflows, interface specifications, or exact Phase 1 implementation boundaries. Therefore, the integration requirements are **TBD** and require technical clarification.

No specific integration mechanism — Microsoft Graph API, SMTP, webhook architecture, a Teams bot framework, or otherwise — is assumed or invented. Until clarified, this is phrased as: “**Scope identified; implementation details and Phase 1 boundary TBD,**” not “Future Phase.”

23. Architecture

This entire section is a PROPOSED TECHNICAL SOLUTION. None of the architecture below was specified by Renewsys; it represents the implementation team’s recommended approach to satisfying the confirmed requirement in Section 9.1.

23.1 Conceptual Flow (Proposed)

User

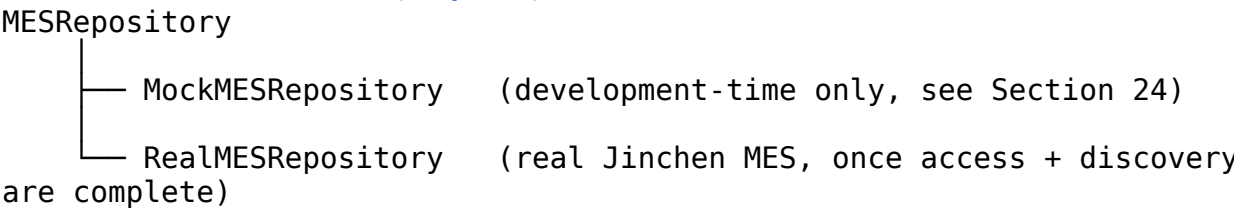
- Web/Chat UI
- Authentication / RBAC
- Application/API Layer – FastAPI or equivalent (Proposed)
- Conversation / session context
- Intent / semantic interpretation
- Schema / metadata layer
- Natural-language-to-SQL generation
- SQL safety review (confirmed: SELECT-only + read-only account; proposed: additional validation/limits – see Section 13.3)
- Read-only MES database (engine TBD – accessed via an adapter)
- Result processing
- KPI / calculation layer

- Visualization layer
- Response to user

On FastAPI specifically: G1 states the planned application-server *location* is “same company LAN / server room,” using the term “FastAPI” in the question text itself. This does **not** constitute a stakeholder mandate for the FastAPI framework. In all diagrams and requirement text, this is labeled:

Application/API Layer — FastAPI (Proposed), or FastAPI or equivalent (Proposed)

23.2 Data-Access Abstraction (Proposed)



23.3 Stakeholder-Confirmed Constraints vs. Proposed Technology

Stakeholder-confirmed constraints	Proposed technology (not stakeholder-approved)
Internal company network deployment (G1, G4)	FastAPI or equivalent application framework
No external cloud AI (K2)	Appropriate frontend framework
Local application accounts (G5)	Database adapter layer (engine TBD)
RBAC (G6)	SQL parser/validator (proposed safety control)
Read-only database access, SELECT-only, dedicated account (F1, F2, F6)	Local/private LLM deployment (specific vendor/model TBD)
English language (K4)	Vector/database components, if required
Session memory (K5)	Charting library

Technology selections in the right-hand column remain **PROPOSED** until validated by IT/security/architecture review. No approved vendor list, mandatory architecture, or prohibited technology list is confirmed — see Section 33 (Constraints).

24. Conceptual Mock Data Contract

CONCEPTUAL / MOCK ONLY — NOT THE ACTUAL JINCHEN MES SCHEMA.

To allow development to begin before the real Jincheng MES schema is confirmed, the implementation team may use a development-time mock data model. Any mock entities, fields, or relationships (e.g., Production, ProcessEvent, Quality, Scrap, WorkOrder, Material, Line,

Shift, WIP, Downtime) must be visibly and permanently labeled as conceptual/mock, and must **never** be presented as discovered Jinchén tables (e.g., a name such as Jinchén.Production must never appear unless independently verified against the real schema).

Production mapping shall only be created after (a) authorized, read-only schema discovery of the real Jinchén MES, and (b) business validation of what each discovered field actually means (Section 20). Mock sample data carries no evidentiary value about the real MES's actual structure or content.

Two-Track Delivery (Proposed)

Track A — Application Development: using the mock data contract, develop chat UI, session memory, intent understanding, query generation, SQL safety controls, result formatting, tables/KPI cards/charts, drill-downs, export, RBAC, login/logout handling, and error handling.

Track B — MES Discovery & Validation: in parallel, establish the actual DB engine, schema, tables/views, relationships, data types, business definitions, KPI formulas, data quality, source-of-truth ownership, network connectivity, security permissions, and performance characteristics.

The mock schema used in Track A is never presented as the production schema.

25. Error Handling

Both error-message fields were left blank by the stakeholder — genuinely unanswered, not “confirmed as none”:

- **U5 (“What should the user see when the database is unavailable?”)** — No response provided. **TBD.**
- **U6 (“What should the user see when the AI cannot answer confidently?”)** — No response provided. **TBD.**

Recommended wording may be proposed, clearly labeled **PROPOSED TECHNICAL SOLUTION**, and never presented as stakeholder-approved:

DB unavailable (Proposed): *“The MES data source is currently unavailable. Please try again later or contact IT.”* AI confidence (Proposed): *“I could not determine a sufficiently reliable answer from the available MES data. Please clarify your question or contact the responsible team.”*

Related confirmed constraints: ask for clarification rather than guess (U1, CONFIRMED); do not always refuse when data is unavailable (U2, CONFIRMED as a negative constraint — see Section 11).

26. Performance

Preserve stakeholder responses without reinterpretation:

Metric	Stakeholder Response	Interpretation	Status
Concurrency	3/4	Ambiguous exact requirement (approximate concurrent-user count)	TBD (precise target)
Peak questions	10/15	Unit/time interval unspecified	TBD
Normal response	Under 5/10 second	Ambiguous target	TBD
Complex response	30 second	Stakeholder-provided; exact scope (max/average/percentile) TBD	Requires validation
Availability	24x7	Stakeholder-provided	Requires operational clarification (see Section 27)

No value in this table is reinterpreted into a precise figure. **10/15 remains ambiguous** (unit/time interval undetermined). **“Under 5/10 second” remains ambiguous** (whether the target is 5s, 10s, or a 5–10s band, and whether it is an average, percentile, or maximum, is unspecified). **30 seconds is preserved as stakeholder-provided**, with its exact measurement basis flagged as requiring clarification.

Database size (R3) and daily transaction volume (R4): **no response provided — TBD.**

27. Availability

R7 = 24×7 (CONFIRMED as stated). S1 = High availability required? “No” (CONFIRMED as stated).

These two responses appear potentially inconsistent and are treated as an **open issue**, not silently resolved by introducing an HA architecture on Renewsys’s behalf. IT must clarify:

- Required uptime (e.g., percentage, or simply “intended usage window”)
- Maintenance windows
- Planned downtime
- Recovery expectations
- Whether “24×7” means guaranteed operational availability or merely the intended usage window during which the system should be accessible

Backup/disaster recovery (S2): no response provided — **TBD**. No backup frequency, RTO, or RPO is invented.

28. Logging and Audit

Item	Response	Classification
What is logged	User login/logout (T1)	CONFIRMED
Audit logging of every AI question and generated SQL	No (G8)	CONFIRMED — not required
Log/audit retention	Not required (T2)	CONFIRMED
Sensitive-value masking in logs	Yes (T3)	CONFIRMED
Monitoring requirements	No response provided (S4)	TBD

Important distinction: “Audit every question/generated SQL = No” (G8) must **not** be read as “no logging at all.” The following are kept distinct:

- **Login/logout logging** — CONFIRMED required (T1).
- **Application/operational logging** (e.g., errors, uptime) — not addressed by the questionnaire; any such logging is a **PROPOSED TECHNICAL SOLUTION** if introduced by the architecture, and is not a business audit-logging requirement.
- **Query/SQL audit logging** — CONFIRMED **not required** (G8).
- **SQL visibility to users** — a separate, RBAC-governed UI concept (I3), not a logging concept.
- **Audit retention** — CONFIRMED not required (T2).

Sensitive-field identification and masking rules require confirmation from IT/security/data owners; no specific field (personal or confidential) is invented here.

29. Data Quality

C6 (“Known data-quality issues”) = “All. Sometimes.” Classification: AMBIGUOUS. This does not constitute a complete data-quality issue inventory.

The **only specifically identified data-quality issue** is (D8, CONFIRMED): “*The data sometimes shows same entries as the scanner scan[s] 1 barcode twice*” (duplicate barcode scanner scans). Other data-quality issues remain **TBD** and are not invented.

The exact deduplication rule (identification key, time window, whether duplicates are removed per-station or globally, which record is retained) is **not specified** anywhere in the questionnaire and remains TBD, along with the broader “IDK” items in Section 20 (D10–D13).

30. UAT

30.1 Confirmed Stakeholder UAT Examples (V2, preserved verbatim, grammar lightly normalized)

- “How many models are produced in line one?”
- “How many scrap models are generated in this month?”
- “How many models are generated in one week?”
- “From which line has this module been passed?”
- “How much profit do we get in this month?”

30.2 UAT Scenario Table with Dependencies

Test ID	Question	Required Data Dependency	Status / Note
UAT-01	“How many models are produced in line one?”	Validated production-event definition, scoped to KM1	Depends on schema discovery + business validation (Section 20)
UAT-02	“How many scrap models are generated in this month?”	Validated Scrap definition (D7), confirmed timezone (L5)	Depends on Section 20/27 open items
UAT-03	“How many models are generated in one week?”	Ambiguous “week” reference — expected to trigger clarification (U1)	Depends on FR-CORE-004
UAT-04	“From which line has this module been passed?”	Traceability data; resolution of duplicate-scan handling (D8, D13)	Depends on data-quality validation
UAT-05	“How much profit do we get in this month?”	Dependency: Financial data source not yet confirmed.	Blocked — see Sections 21, 30.3
UAT-06	“What was today’s production?”	Validated production-event definition	Depends on schema/business validation
UAT-07	Database temporarily unreachable	Error-message wording (U5, TBD)	Depends on Section 25
UAT-08	Non-admin user requests to see SQL	RBAC enforcement (I3)	Depends on Section 18

Test ID	Question	Required Data Dependency	Status / Note
UAT-09	Follow-up question omitting prior scope	Session memory behavior (K5)	Depends on FR-CORE-003
UAT-10	Export a query result	Export format decision (I2)	Depends on Section 16

30.3 Profit Question — Explicit Treatment

Profit calculation requires validated financial data sources such as revenue, cost, or margin information. The questionnaire does not establish that these data are available in Jincheng MES. Therefore the chatbot shall not fabricate a profit value.

This is documented as a cross-system/data-source dependency requiring confirmation, not a capability the system can be expected to pass UAT-05 against until that source is available.

No numeric expected answer is fabricated for any UAT scenario.

31. Non-Functional Requirements

Category	Confirmed Requirement	Proposed Technical Control	TBD
Security	Read-only DB access, SELECT-only, dedicated read-only account (F1/F2/F6); RBAC (G6); no external cloud AI (K2); local application accounts (G5); sensitive-value masking (T3)	SQL parser/validator, additional query limits (Section 13.3)	Compliance/security standards (T4)
Authentication	Local application accounts (G5)	—	User provisioning process
Performance	— (all figures ambiguous, see Section 26)	Load-testing plan once units are clarified	Concurrency, peak load, response-time targets
Availability	24×7 stated (R7); HA not required (S1)	—	Reconciliation of the 24×7/HA tension (Section 27); backup/DR
Scalability	—	Adapter-based architecture to allow future scaling	Infrastructure sizing

Category	Confirmed Requirement	Proposed Technical Control	TBD
		(Proposed)	
Logging	Login/logout logging (T1); no mandatory query audit (G8); no retention requirement (T2)	Operational/error logging (Proposed)	Monitoring requirements (S4)
Data Masking	Sensitive values masked in logs (T3)	—	Specific sensitive fields
Maintainability	—	Adapter-based DB layer, configurable RBAC (Proposed)	—
Observability	—	Proposed if architecture requires it	Monitoring tooling (S4)

32. Assumptions

Every assumption is explicitly labeled and never hidden inside a requirement statement.

ID	Assumption	Reason	Risk if Wrong	Validation Owner
AS-01	An adapter-based database access layer (FastAPI or equivalent application layer) is a workable implementation approach.	B1 = “Over to you”; database engine unknown.	Development approach may need to change once the real engine is confirmed.	IT/DBA
AS-02	A SQL validation/safety layer beyond SELECT-only + read-only account can be added without stakeholder objection.	U3 = “No” for explicit validation, but layered safety is good practice for a production read-only system.	May be seen as scope the stakeholder did not ask for; requires sign-off before build.	Security/Architecture

ID	Assumption	Reason	Risk if Wrong	Validation Owner
AS-03	The conceptual mock schema (Section 24) is a reasonable stand-in for early development.	No real schema is available (C1).	Mock assumptions could diverge significantly from the real Jinchen schema, requiring rework.	Development Team
AS-04	A locally or privately hosted LLM architecture is achievable to satisfy the no-external-cloud-AI requirement.	K2 = data must stay inside company network; no vendor/model decided (K1, K3).	If no compliant hosting option exists, the natural-language layer cannot be built as envisioned.	IT/Security
AS-05	Standard industry KPI formulas (Section 14.2) are a reasonable starting point pending validation.	M2 = “All,” not actual formulas.	Formulas may not match Renewsys’s actual business definitions once validated.	Business/MES SME
AS-06	The MES database can be reached from the planned application environment once network/firewall details are clarified.	B7 = “Yes” (stated, not network-tested); G3 = firewall “need to know.”	Application cannot connect; deployment delay.	IT/Network
AS-07	Authorized, read-only database access can eventually be provisioned even though the respondent cannot describe the schema	B10, C1 — access presumed obtainable via formal request.	Development may be blocked indefinitely on schema/data access.	IT/DBA/MES vendor

ID	Assumption	Reason	Risk if Wrong	Validation Owner
	themselves.			

33. Constraints

- **V4 (deployment architecture constraints) = “Idk.” Classification: AMBIGUOUS.** No mandatory architecture (containerization, cloud vs. on-premises, specific server specification) is confirmed.
- **V5 (approved technology constraints) = “Anything which is good.”** No approved vendor list is confirmed.
- **V6 (technologies not allowed) = “No.”** No prohibited technology list is confirmed.

Therefore: no mandatory architecture, no approved vendor list, and no prohibited technology list are stakeholder-confirmed. Proposed technologies may be recommended based on security, deployment, maintainability, and compatibility considerations (see Section 23.3), but must never be presented as stakeholder-approved.

34. Risks

Risk	Impact	Likelihood	Mitigation	Owner
Unknown database engine	High — affects driver, tooling, SQL dialect decisions	Not assessed — insufficient basis to assign	Adapter-based data-access design	IT/DBA
Unknown Jinchen schema	High — blocks finalization of Text-to-SQL logic	Not assessed	Controlled read-only discovery + business validation	IT/DBA/MES vendor
Unknown business definitions (Material, quantity semantics, process loss, rework)	High — affects multiple KPIs and question categories	Not assessed	Structured business-validation workflow	Business/MES SME
Unknown KPI formulas	High — risk of delivering KPIs the business does not accept	Not assessed	KPI validation table + sign-off (Section 14)	Business/MES SME

Risk	Impact	Likelihood	Mitigation	Owner
Unknown KPI source-of-truth ownership	Medium — ambiguity in who “owns” a number	Not assessed	Source-of-Truth Register (Section 21)	Business/IT
Network-location ambiguity (MES server “in China” vs. planned company-LAN application)	Medium/High — could block connectivity entirely	Not assessed	Explicit clarification with IT/MES vendor (Section 12.4)	IT/MES vendor
Vendor-controlled MES (Jinchen)	Medium — limited local visibility/control over schema and connection mechanism	Not assessed	Formal engagement with vendor/IT for schema and API detail	IT/MES vendor
Data-quality issues (duplicate scanner scans and possibly others)	High — affects count accuracy across many KPIs	Not assessed	Data-quality profiling once real data access exists	Business/MES SME
Performance ambiguity (10/15, “under 5/10 second”)	Medium — risk of building to the wrong target	Not assessed	Clarify units/measurement basis before load testing	Business/IT
24x7 vs. HA ambiguity	Medium — unclear uptime commitment	Not assessed	IT clarification of uptime/maintenance expectations	IT
Missing backup/DR requirements	Medium	Not assessed	Obtain requirements from IT before production go-live	IT
Missing monitoring requirements	Low/Medium	Not assessed	Obtain requirements from IT	IT
Missing compliance/security standards	Medium	Not assessed	Confirm applicable standards with	IT/Security

Risk	Impact	Likelihood	Mitigation	Owner
Email/Teams integration uncertainty	Medium — scope named but undefined	Not assessed	IT/security Obtain API/workflow detail (Section 22)	IT/Business
Financial/profit data dependency	Medium — UAT-05 cannot pass without it	Not assessed	Confirm financial data source before implementing any profit-related answer	Finance/IT

Likelihood is marked “**Not assessed**” throughout because the questionnaire and project team have not supplied likelihood/probability estimates; none are fabricated here.

35. Open Issues / Clarifications

1. Actual database engine/technology (B1)
2. Actual Jinchen MES schema (C1)
3. Database network location and topology (B6 vs. G1/G2 — see Section 12.4)
4. Precise meaning of “MES server is in China” in relation to hosting/ownership (B6)
5. Reachability of 10.69.12.20 from the intended application server, and required network path (B7, G2)
6. Firewall/network restrictions (G3)
7. Database authentication mechanism to be used by the application (B5 — SQL vs. AD)
8. Detail of SQL vs. AD authentication configuration
9. Exact RBAC role-to-data-domain mapping (G7, J2)
10. Actual user/role provisioning process
11. Email integration — APIs, authentication, workflow (Q1/Q2)
12. Microsoft Teams integration — APIs, authentication, workflow (Q1/Q2)
13. KPI formulas (M2)
14. KPI ownership / source of truth (M3)
15. KPI thresholds/targets (M4)
16. MES timezone (L5)
17. Quantity semantics — cumulative vs. transactional (D10)
18. Material definition (D9)
19. Process-loss definition (D11)
20. Rework-loop definition (D12)
21. Duplicate/repeated event handling beyond the confirmed barcode-scan issue (D13)
22. Full inventory of data-quality issues beyond duplicate scans (C6)

23. Database size (R3)
24. Daily transaction volume (R4)
25. Peak-question unit/time interval (R2)
26. Normal response-time target and measurement basis (R5)
27. Availability/maintenance interpretation — 24×7 vs. HA=No (R7, S1)
28. Backup/disaster recovery requirements (S2)
29. Monitoring requirements (S4)
30. Compliance/security standards (T4)
31. Database-unavailable message wording (U5)
32. AI low-confidence message wording (U6)
33. Financial/profit data source (Q3, V2)

Additional items surfaced during the field-by-field audit (Section 38): named individual contacts beyond “Managers, HOD” (A9); specific measurable business outcomes beyond “All” (A5); whether MES retains data beyond 30 days even though chatbot querying is capped there (L2); exact export formats (I2); exact branding guidelines (I5); organizational SRS/UAT approval process beyond the named “Developer” (V3); and a formal follow-up list with IT/DBA/MES vendor (FNL2 was answered “None” despite FNL1 acknowledging assumptions exist).

36. Deployment

Item	Response	Classification
Application server location	Same company LAN / server room (G1)	CONFIRMED
Network path to MES DB	Direct LAN (G2)	CONFIRMED
Operator/maintainer	IT (S3)	CONFIRMED
Required environments	“All” (S5)	AMBIGUOUS — not interpreted as a specific Dev/Test/UAT/Prod topology without clarification
Deployment architecture constraints	“Idk” (V4)	AMBIGUOUS/TBD

No specific environment list, containerization approach, or infrastructure topology is invented.

37. Future Extensions / TBD Roadmap

V7 (“Future features planned after Version 1”) is blank.

Future roadmap items have not been explicitly provided in the questionnaire. Potential extensions may be identified as TBD/proposed opportunities but must not be presented as confirmed stakeholder commitments.

The only item the questionnaire explicitly supports as deferred is:

- **Company document / RAG functionality (H1: “Future phase”)** — document types, storage location, permission model, and retention are all TBD for this future phase (H2–H5).

Email, Microsoft Teams, and advanced predictive analytics are explicitly NOT placed in this section — see Sections 22 and 15 respectively for their correct (in-scope-but-TBD / proposed-TBD) treatment.

Transactional MES operations are noted in F1 as something that “may be added later,” but this is not a committed roadmap item and is not restated here as one.

38. Requirements Traceability Matrix — Complete 139-Field Matrix

Every one of the 139 fields in the questionnaire is represented exactly once below — including answered, ambiguous (“All”, “Over to you”, “IDK”, “Need to request”), and blank/content-free fields. No field is described as a “representative sample”; this is the complete matrix.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
META-1	2026-09-25 21:17:17.459000	Administrative — records when the questionnaire was submitted; not a requirement.	N/A — Administrative	Recorded for document-control purposes only; not a requirement.	1. Document Control	None — retained for record only.
META-2	samikshapalkar20@gmail.com	Administrative — identifies the respondent; not a functional requirement.	N/A — Administrative	Recorded for document-control purposes only; not a requirement.	1. Document Control	TBD — confirm respondent’s name/role for the stakeholder register.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		nt. Full name/role of respondent not captured.				
A1	Renewsys Chatbot	Confirms the project name as stated by the stakeholder.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	3. Executive Summary / Business Problem	Preserve as-is.
A2	Renewsys India Pvt Ltd	Confirms customer/company identity.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	1. Document Control	Preserve as-is.
A3	Khopoli	Confirms the Phase 1 site.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	7. Scope	Preserve as-is.
A4	Instead of retrieving data directly from the database we can take help from this chat what it can translate from plane English to the queries and	Confirms the core business problem: replace manual DB/report retrieval with natural-language chat.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	4. Business Problem	Preserve verbatim intent; do not add invented outcomes.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	generate the required fields					
A5	All	Vague — does not specify measurable outcomes (e.g., time saved, cost reduced).	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	5. Objectives and Outcomes	Clarification required — request specific, measurable success criteria from stakeholders.
A6	Plant Manager, Production Manager, Quality Manager, Production Planning, Management / Director, IT / System Administrator	Confirms the business roles who will use the system.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	6. Stakeholders and Users	Preserve as-is.
A7	15/20	Approximate total user count expressed as a range; treated as an estimate rather than an exact	CONFIRMED	Preserved in the SRS as a confirmed requirement.	6. Stakeholders and Users	Preserve as approximate figure (~15–20 users).

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
A8	Single plant	figure. Confirms Version 1 is limited to a single plant (Khopoli, per A3).	CONFIRMED	Preserved in the SRS as a confirmed requirement.	7. Scope	Preserve as-is.
A9	Managers, HOD	Names roles only ('Managers, HOD'); no specific individual names or responsibilities were provided.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	6. Stakeholders and Users	TBD — obtain named contacts and their specific responsibilities.
B1	Over to you	Stakeholder explicitly delegates the choice/knowledge of DB technology to the implementation team; does NOT confirm an engine.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	11. MES/Data Requirements	Do not select an engine (e.g., SQL Server/MySQL/Oracle) without IT/DBA confirmation. Mark engine as TBD.
B2	10.69.12.20	Specific technical value provided by the stakeholder; connectivity	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/M	11. MES/Data Requirements	Verify reachability and correctness with IT; treat as sensitive information

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		y/correctness not independently verified.	ION	ES vendor.		n — restrict its use in documentation.
B3	8000.0	Specific technical value provided by the stakeholder; not independently verified.	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	11. MES/Data Requirements	Verify with IT/DBA.
B4	Jinchen MES	Confirms the name of the MES/data base as known to the stakeholder; internal schema/catalog name not confirmed.	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	11. MES/Data Requirements	Verify actual catalog/database name with IT/DBA during discovery.
B5	Both SQL and AD	Confirms two authentication mechanisms exist for the DB (SQL auth and Active Directory).	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	11. MES/Data Requirements / 21. Security Requirements	Verify with IT which mechanism the application account should use.
B6	Mes server is in China we do not	Confirms the DB/server	STAKEHOLDER-PROVIDED	Retained as provided;	11. MES/Data Requirements	Clarify ownership, SLAs, and

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	handle	is hosted/managed outside of local Renewsys IT control (associated with the Jinchen/China environment).	— REQUIRES TECHNICAL VERIFICATION	requires technical verification by IT/DBA/MES vendor.	nts	support contact for the hosting party.
B7	Yes	Stakeholder states reachability is expected; not independently network-tested.	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	21. Network/Deployment Requirements	Verify via network test once environments are defined.
B8	Yes we can	Confirms read-only/reporting views may exist and can potentially be provided; exact views not identified.	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	11. MES/Data Requirements	Request the specific list of views from IT/DBA.
B9	I don't know	Stakeholder explicitly does not know if stored procedures	AMBIGUOUS	Preserved as ambiguous stakeholder input; not	11. MES/Data Requirements	TBD — request from IT/DBA/MES vendor.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		/APIs exist.		converted into a specific requirement.		
B10	I don't know we are not developers it was made by Chinese jinchen server	Stakeholder confirms they are not developers and do not know the existing connection architecture.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	11. MES/Data Requirements	TBD — request from MES vendor (Jinchen) or IT/DBA.
C1	Need to request from IT / DBA	Confirms the schema is not currently in the respondent's possession and must be formally requested.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	11. MES/Data Requirements	Action: formally request schema from IT/DBA before production Text-to-SQL design is finalized.
C2	Data types	Confirms only data types are currently expected to be available; table/column names, keys, and relationships are not confirmed	CONFIRMED	Preserved in the SRS as a confirmed requirement.	11. MES/Data Requirements	Request full schema (tables, columns, keys, relationships) in addition to data types.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
C3	All	as available. Vague — does not name specific tables/views.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	11. MES/Data Requirements	Clarification required — request the actual list of tables/views.
C4	It is understood	Vague — indicates the relationships exist within MES but does not describe them.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	Clarification required — relationships must be obtained via schema discovery and business validation.
C5	Partially	Confirms the sample report only partially reflects the real DB structure; extent of divergence unknown.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	11. MES/Data Requirements	Do not assume the sample Excel structure equals the schema; validate during discovery.
C6	All. Sometimes	Vague as a category answer, but the only specifically identified	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted	11. MES/Data Requirements	Clarification required for issues beyond duplicate scans; do not invent

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		issue elsewhere is duplicate barcode scans (see D8).		into a specific requirement.		additional data-quality problems.
D1	A morning B second C night	Confirms mapping: A=Morning, B=Second, C=Night.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve; validate against actual MES ShiftName column values during discovery.
D2	Under Mes	Vague — confirms only that this is an MES-defined process concept; exact values not provided.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	TBD — obtain actual TechnologyStepName values from MES during discovery.
D3	They are module activities	Confirms ACTIVITY relates to module activities at a general level; exact code list not provided.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	TBD — obtain actual ACTIVITY code list.
D4	Module in that station	Business-level definition:	CONFIRMED	Preserved in the SRS as a	12. Business Semantics	Preserve; validate exact DB

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		a module entering a station.		confirmed requirement.		representation during discovery.
D5	Out from station	Business-level definition: a module leaving a station.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve; validate exact DB representation during discovery.
D6	Module have defect	Business-level definition: a module with a defect.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve; validate exact DB representation during discovery.
D7	Damage in scrap	Business-level definition: a damaged/s crapped module.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve; validate exact DB representation during discovery.
D8	The data sometimes shows same enteries as the scanner scan 1 barcode twics	Confirms the specific data-quality issue: the scanner may scan the same barcode twice, producing duplicate entries.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics / 13. NL Query Requirements	Preserve as the only confirmed data-quality issue; deduplication algorithm remains TBD (see Section 13/61).
D9	Area-location, Workorder - order which we get to	Confirms business-level definitions for Area, Workorder	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve definitions verbatim; Material remains AMBIGUO

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	produce, Material-idk, Start Time- date and time of report, End Time- time and date of report which required, Shift-ABC shifts which report required and Line we have 3 lines KM1, KM2 and KM3	, Start Time, End Time, Shift, and Line. Material is explicitly unknown.				US/TBD — no meaning invented.
D10	Idk	Stakeholder explicitly does not know.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	TBD — requires business/ MES validation before KPI/Text-to-SQL logic assumes either interpretation.
D11	Idk	Stakeholder explicitly does not know.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a	15. KPI Requirements	TBD — no process-loss formula may be invented; requires business

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
				specific requirement.		validation.
D12	Idk	Stakeholder explicitly does not know.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	TBD — requires MES process validation.
D13	Idk	Stakeholder explicitly does not know (separate from the specifically confirmed barcode-duplicate issue in D8).	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	12. Business Semantics	TBD — requires data profiling/validation; do not assume scope beyond D8.
E1	“What was today’s production?”, “How many units entered Glass?”, “How many units are in Packing? From which line does this module is	Confirms representative example questions (fewer than 10 distinct examples were actually given, despite the request for 10+).	CONFIRMED	Preserved in the SRS as a confirmed requirement.	13. Natural-Language Query Requirements	Preserve examples verbatim (grammar normalized only for readability); treat as illustrative, not exhaustive.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	produce? At which time the module produce? How much module create on this days? How many module produces on this line? How many scrap/bad found? How many production happen on this day?					
E2	All	Vague — indicates broad support desired but names no specific questions.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Clarification required — collect concrete example questions during UAT preparation.
E3	All	Vague — same pattern as E2.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Clarification required.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
E4	All	Vague — same pattern as E2.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Clarification required.
E5	All	Vague — same pattern as E2.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Clarification required.
E6	All	Vague — same pattern as E2.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Clarification required.
E7	Over to you	Stakeholder delegates this decision to the implementation/desi	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted	13. Natural-Language Query Requirements	A safe-answering policy may be proposed (Section 16 of master

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		gn team.		into a specific requirement.		prompt) but must be labeled Proposed, not stakeholder-confirmed.
E8	Over to you	Stakeholder delegates specifics to the implementation team; the general principle (clarify rather than guess) is separately confirmed at U1.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	13. Natural-Language Query Requirements	Propose clarification triggers (Section 16); label Proposed.
F1	Read-only now; transactional actions may be added later	Confirms Phase 1 is read-only; write/transactional capability is explicitly deferred, not committed.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	9. Functional Requirements / 8. Out of Scope	Preserve; treat write operations as Future Phase, not Phase 1.
F2	SELECT only	Confirms SELECT-only operation.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	14. SQL Generation and Safety	Preserve as a confirmed security control.
F3	No	Confirms no write	CONFIRMED	Preserved in the SRS	8. Out of Scope	Preserve.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
F4	Fully accessible	operations are required in Phase 1. Confirms no specific access restriction was named by the stakeholder; does NOT mean the chatbot should indiscriminately query every object without controls.	AMBIGUOUS	as a confirmed requirement. Preserved as ambiguous stakeholder input; not converted into a specific requirement.	14. SQL Generation and Safety	Apply least-privilege/approved-object principles regardless; do not equate 'fully accessible' with unrestricted automatic querying (see Section 17 caution).
F5	No	Confirms no specific exclusions were identified by the stakeholder.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	14. SQL Generation and Safety	Preserve; revisit once schema is discovered in case sensitive fields are found.
F6	Yes	Confirms a dedicated read-only account is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	14. SQL Generation and Safety	Preserve as a confirmed security control.
G1	Same company LAN /	Confirms intended internal	CONFIRMED	Preserved in the SRS as a	21. Network/Deployment	Preserve deployment intent; do

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	server room	deployment location. Note: 'FastAPI' appears in the question wording itself, not as a stakeholder-mandated technology.		confirmed requirement.	t Requirements	not treat 'FastAPI' framework naming as a confirmed technology mandate beyond internal deployment.
G2	Direct LAN	Confirms intended direct LAN connectivity.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	21. Network/Deployment Requirements	Verify actual network path/firewall rules with IT.
G3	Need to known	Stakeholder does not currently know firewall/network restrictions.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	21. Network/Deployment Requirements	TBD — obtain from IT/network team; do not invent ports or topology.
G4	No	Confirms public internet exposure is not permitted.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	21. Network/Deployment Requirements / 20. Security Requirements	Preserve as a confirmed security constraint.

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G5	Local application accounts	Confirms local application accounts, not AD/SSO, for chatbot login.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	20. Security Requirements	Preserve; do not substitute AD/SSO unless separately approved.
G6	Yes	Confirms RBAC is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	10. User Roles and Access	Preserve.
G7	On the mes developooer not on us	Confirms detailed role/data-visibility mapping is owned by the MES developer side, not the respondent, and is therefore not currently available.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	10. User Roles and Access	TBD — obtain detailed role-permission mapping from MES developer/IT.
G8	No	Confirms full question/S QL audit logging is not required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	24. Logging/Audit	Preserve; do not introduce mandatory full audit logging.
G9	None	Confirms no specific retention requirement	CONFIRMED	Preserved in the SRS as a confirmed	24. Logging/Audit	Preserve as 'not required'; revisit if

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		nt was identified.		requirement.		compliance requirements (T4) are later defined.
H1	Future phase	Confirms document/RAG capability is explicitly deferred beyond Phase 1.	FUTURE PHASE	Retained as an explicitly stated future-phase item; excluded from Phase 1 functional scope.	34. Future Phase	Exclude from Phase 1 functional requirements.
H2	None	No response provided beyond 'None' — no information provided; consistent with H1 being future phase.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	34. Future Phase	TBD for future phase scoping.
H3	None	No response provided beyond 'None' — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	34. Future Phase	TBD for future phase scoping.
H4	Unknown / To be confirmed	Explicitly unresolved .	TBD / TO BE CONFIRMED	Marked TBD in the SRS;	34. Future Phase	TBD for future phase

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
			ED	requires confirmation before implementation.		scoping.
H5	None	No response provided beyond 'None' — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	34. Future Phase	TBD for future phase scoping.
I1	Natural-language answer, Tables, KPI cards, Bar charts, Line charts, Pie / donut charts, Trend charts, Drill-down details, Source / query details for authorized users	Confirms the full list of required response/output formats.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	17. Visualization Requirements	Preserve list verbatim.
I2	All	Confirms export capability is desired in general; does not specify exact formats.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific	17. Visualization Requirements	Do not assume specific formats (CSV/Excel/PDF) are confirmed; propose formats

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				requirement.		and label Proposed.
I3	Only administrators / IT	Confirms SQL/source visibility is restricted to Admin/IT role.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	10. User Roles and Access / 17. Visualization Requirements	Preserve.
I4	Yes	Confirms freshness/timestamp display is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	27. Data Freshness/History	Preserve.
I5	Yes	Confirms branding/UI requirements exist; no detailed design was provided.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	17. Visualization Requirements	TBD — obtain specific branding guidelines/assets.
J1	Admin and users	Confirms two platform-level roles.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	10. User Roles and Access	Preserve.
J2	All	Vague — does not define per-role restrictions.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a	10. User Roles and Access	TBD — detailed permission mapping required (also see G7).

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
J3	Yes	Confirms this capability is required; the identity/attribute source mechanism is not specified.	CONFIRMED	specific requirement. Preserved in the SRS as a confirmed requirement.	10. User Roles and Access	Preserve requirement; source-of-attribute mechanism is TBD.
J4	No	Confirms row-level security is NOT required in Phase 1.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	10. User Roles and Access	Preserve; do not introduce row-level security as a Phase 1 requirement.
K1	Not decided	Confirms no LLM/vendor decision has been made.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	26. AI/LLM Requirements	Do not select a vendor/model; mark as TBD pending decision.
K2	No — data must stay inside company network	Confirms external cloud AI processing is prohibited.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	26. AI/LLM Requirements / 20. Security Requirements	Preserve as a confirmed, non-negotiable constraint.
K3	No	Confirms no approved vendor/model	CONFIRMED	Preserved in the SRS as a confirmed	26. AI/LLM Requirements	Preserve; vendor selection is TBD/Propo

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		del exists yet.		requirement.		sed pending IT/security approval.
K4	English	Confirms English-only for Phase 1.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	26. AI/LLM Requirements	Preserve.
K5	Yes	Confirms session-scoped conversational memory is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	18. Session/Conversation Requirements	Preserve; do not assume persistent cross-session memory.
L1	In 24 hours	Confirms freshness expectation of approximately 24 hours.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	27. Data Freshness/History	Preserve; do not claim real-time/streaming freshness.
L2	30 days	Confirms the chatbot's query scope is limited to 30 days.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	27. Data Freshness/History	Preserve; clarify separately whether older MES data exists but is simply excluded from chatbot querying (TBD).
L3	Yes	Confirms this capability is required.	CONFIRMED	Preserved in the SRS as a confirmed	13. Natural-Language Query	Preserve.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
L4	Today, Yesterday, Current shift, Previous shift, Last 7 days, Last 30 days, This week, Last month, Custom date range, Custom time range	Confirms the specific list of required date/time expressions.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	13. Natural-Language Query Requirements	Preserve list verbatim.
L5	We dont know	Stakeholder explicitly does not know the MES timezone.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	27. Data Freshness/History	TBD — must be verified with IT/DBA; do not assume IST or UTC.
L6	7-3 morning 3-11 second 11-7 night	Confirms shift timing: Morning 07:00–15:00, Second 15:00–23:00, Night 23:00–07:00.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	12. Business Semantics	Preserve exactly; validate mapping to actual MES ShiftName values and confirm timezone (L5) before relying on this in production

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
M1	Total production, Get-in quantity, Departure quantity, Bad quantity, Scrap quantity, Yield, Defect rate, Scrap rate, Process loss, OEE, Availability, Performance, Quality, Downtime, Cycle time, WIP, Throughput, First Pass Yield, Rework rate	Confirms the list of required KPIs.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	15. KPI Requirements	. Preserve list verbatim as stakeholder-confirmed KPI names.
M2	All	Does NOT actually supply any formula; 'All' cannot be read as formulas having been provided.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	15. KPI Requirements	Clarification required — no KPI formula may be treated as Renewsys-approved without further input.
M3	Over to you	Stakeholder delegates	AMBIGUOUS	Preserved as	15. KPI Requirements	A source-of-truth

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		this decision to the implementation team.		ambiguous stakeholder input; not converted into a specific requirement.	nts	strategy may be proposed but must be labeled Proposed, pending stakeholder/IT confirmation.
M4	Over to you	Stakeholder delegates this decision to the implementation team.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	15. KPI Requirements	Do not invent target values or thresholds; any proposal must be labeled Proposed/TBD.
N1	Day-over-day comparison, Shift comparison, Line comparison, Process comparison, Product / material comparison, Trend analysis, Top loss processes, Top defect processes, Anomaly	Confirms the list of required analytical capabilities.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	16. Analytics Requirements	Preserve list verbatim; note several capabilities depend on unresolved business semantics (D10–D12).

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	detection, Root-cause investigation support, What-if calculations					
N2	Over to you	Stakeholder delegates specifics to the implementation team.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	16. Analytics Requirements	Propose example scenarios; label Proposed.
N3	Yes	Confirms this distinction is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	16. Analytics Requirements	Preserve as a confirmed UX/output requirement.
N4	Yes — informational only	Confirms recommendations/predictions are allowed, explicitly informational only (not operational/transactional).	CONFIRMED	Preserved in the SRS as a confirmed requirement.	16. Analytics Requirements	Preserve; do not present as operational commands or guarantees.
O1	No	Confirms alerts are not required in	CONFIRMED	Preserved in the SRS as a confirmed	8. Out of Scope	Preserve — Out of Scope.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		Phase 1.		requirement.		
O2	None	Consistent with O1 = No.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	No action — Out of Scope.
O3	None	Consistent with O1 = No.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	No action — Out of Scope.
O4	None	Consistent with O1 = No.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	No action — Out of Scope.
P1	No	Confirms scheduled reports are not required in Phase 1.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	Preserve — Out of Scope.
P2	None	Consistent with P1 = No.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	No action — Out of Scope.
P3	None	Consistent with P1 = No.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	8. Out of Scope	No action — Out of Scope.
Q1	MES, Email,	Confirms three	CONFIRMED	Preserved in the SRS	19. Integration	Preserve MES,

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	Microsoft Teams	systems are named as in scope. Does NOT by itself confirm specific Email/Teams integration functionality — see Q2.		as a confirmed requirement.	Requirements	Email, and Microsoft Teams as in-scope systems; do not silently move Email/Teams to future phase.
Q2	None	No response provided beyond 'None' — no integration API/interface detail was provided for any of the systems named in Q1.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	19. Integration Requirements	TBD — Email and Microsoft Teams integration requirements, APIs, authentication, and workflows require clarification; do not invent Microsoft Graph API, SMTP, or bot architecture.
Q3	Not knowing	Stakeholder explicitly does not know the source of truth per domain.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific	11. MES/Data Requirements	TBD — establish a Source-of-Truth Validation Register per domain; do not assume

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				requirement.		MES is authoritative for all domains (notably financial/profit).
R1	3/4	Approximate concurrency figure expressed as a range.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	22. Performance Requirements	Preserve as approximate (~3–4 concurrent users).
R2	10/15	Ambiguous — the unit (per minute vs. per hour) and measurement period are not specified in the response.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	22. Performance Requirements	Clarification required — confirm unit and measurement window before using this as a load-testing target.
R3	None	No response provided beyond ‘None’ — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	22. Performance Requirements	TBD — obtain from IT/DBA.
R4	None	No response provided beyond ‘None’ — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	22. Performance Requirements	TBD — obtain from IT/DBA/MES vendor.

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
R5	Under 5/10 second	Ambiguous — could mean ‘under 5 seconds’, ‘under 10 seconds’, or ‘between 5 and 10 seconds’; exact target and measurement definition (average/percentile/max) are unspecified.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	22. Performance Requirements	Clarification required before treating as a firm performance target.
R6	30 second	Stakeholder-provided value; whether this is a maximum, target, average, or percentile is not specified.	STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Retained as provided; requires technical verification by IT/DBA/MES vendor.	22. Performance Requirements	Retain as stakeholder-provided; clarify measurement definition (max/target/average/percentile).
R7	24x7	Confirms a 24x7 availability expectation; no uptime percentage, SLA, or RTO/RPO was	CONFIRMED	Preserved in the SRS as a confirmed requirement.	23. Availability/Operations	Preserve; do not invent an uptime percentage or SLA. Note tension with S1 (see Open

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		specified.				Issues/Conflicts).
S1	No	Confirms high-availability architecture is not required per this response.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	23. Availability/Operations	Preserve; document alongside R7 as a requiring-clarification pairing (24x7 expectation vs. no HA architecture), not silently resolved.
S2	None	No response provided beyond 'None' — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	23. Availability/Operations	TBD — do not invent backup frequency, RTO, or RPO.
S3	IT	Confirms IT as the identified operator.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	23. Availability/Operations	Preserve.
S4	None	No response provided beyond 'None' — no information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	23. Availability/Operations	TBD — do not invent specific monitoring tooling (e.g., Prometheus/Grafana/ELK).
S5	All	Ambiguous — does not	AMBIGUOUS	Preserved as	21. Network/	Clarification required

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		confirm which specific environments (Dev/Test/UAT/Prod) are meant.		ambiguous stakeholder input; not converted into a specific requirement.	Deployment Requirements	before assuming a specific environment topology.
T1	User login/logout	Confirms the scope of required logging.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	24. Logging/Audit	Preserve; do not expand to full question/SQL audit logging (see G8).
T2	Not required	Confirms no specific retention period is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	24. Logging/Audit	Preserve.
T3	Yes	Confirms masking is required; specific sensitive fields not identified.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	24. Logging/Audit	Preserve requirement; specific fields/masking rules remain TBD.
T4	None	No response provided beyond 'None' — no compliance standard was named.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	20. Security Requirements	TBD — do not assume ISO 27001, SOC 2, GDPR, DPDP Act, or any other standard applies unless

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U1	Yes	Confirms ambiguity-clarification behavior is required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	13. Natural-Language Query Requirements	confirmed. Preserve as a confirmed requirement.
U2	No	Confirms the AI should NOT necessarily refuse when data is unavailable; it should not be assumed to always refuse.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	25. Error Handling	Preserve; do not invent a blanket 'refuse when data missing' behavior as confirmed. A transparent partial-answer approach may be proposed and must be labeled Proposed.
U3	No	Confirms the stakeholder did not request an explicit pre-execution SQL validation step.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	14. SQL Generation and Safety	Preserve; do not claim SQL validation was requested by Renewsys. Any validation layer proposed by the implement

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
						ation team must be labeled Proposed Technical Safety Control.
U4	SELECT-only enforcement, Database read-only account	Confirms exactly these two controls as stakeholder-required.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	14. SQL Generation and Safety	Preserve as the only stakeholder-confirmed safety controls; additional controls (parsers, validators, limits) are Proposed.
U5	None	No response provided beyond 'None' — no wording was provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	25. Error Handling	TBD — proposed UX copy may be offered but must be labeled Proposed, not approved.
U6	None	No response provided beyond 'None' — no wording was provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	25. Error Handling	TBD — proposed UX copy may be offered but must be labeled Proposed, not approved.
V1	Chatbot	Confirms	CONFIRM	Preserved	9.	Preserve as

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
	should answer the questions and create required chats, graphs etc by reading the DB .	the central Phase 1 objective: answer questions and produce charts/graphs by reading MES data.	ED	in the SRS as a confirmed requirement.	Functional Requirements	the central Phase 1 objective statement (Section 46 of master prompt).
V2	How many models are produced in line one? How many scrap models are generated in this month? How many models are generated in one week? From which line does this model has been passed? How much profit does we get in this month?	Confirms specific UAT example questions, including a profit question that MES may not be able to answer without a financial data source.	CONFIRMED	Preserved in the SRS as a confirmed requirement.	30. UAT Requirements	Preserve verbatim (grammar normalized only for readability); flag the profit question for data-source validation (Section 47 of master prompt). Do not fabricate expected numeric answers.
V3	Developer	Confirms 'Developer' as the stakeholder	CONFIRMED	Preserved in the SRS as a confirmed	1. Document Control	Preserve exactly as 'Developer' ; do not

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		r-named approver.		requirement.		substitute a different role. Note that broader organizational sign-off may still be advisable.
V4	Idk	Stakeholder explicitly does not know.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	28. Proposed Technical Architecture	TBD — do not invent containerization, cloud provider, or server specifications as confirmed.
V5	Anything which is good	No specific technology is mandated by the stakeholder.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	28. Proposed Technical Architecture	Any technology selected by the implementation team is a Proposed Technical Solution, not a stakeholder mandate.
V6	No	Confirms no specific technology is prohibited by the stakeholder	CONFIRMED	Preserved in the SRS as a confirmed requirement.	28. Proposed Technical Architecture	Preserve — no technology is explicitly excluded by the stakeholder

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
V7	None	No response provided beyond 'None' — no future roadmap was provided by the stakeholder.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	34. Future Phase	Do not invent a future roadmap; list only explicitly-identified future items (e.g., document/RAG per H1).
V8	None	No response provided beyond 'None' — no additional information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	33. Open Issues / Clarifications	No action beyond noting absence of additional input.
V9	None	No response provided beyond 'None' — no additional information provided.	TBD / TO BE CONFIRMED	Marked TBD in the SRS; requires confirmation before implementation.	33. Open Issues / Clarifications	No action beyond noting absence of additional input.
FNL1	Partially confirmed; some items are assumptions	Stakeholder self-assesses the questionnaire as only partially confirmed, explicitly acknowledged	CONFIRMED	Preserved in the SRS as a confirmed requirement.	2. Requirements Baseline Status	This response is the direct basis for the document-level 'Requirements Baseline

Field ID	Questionnaire Response	Interpretation	Classification	SRS Requirement / Treatment	SRS Section	Action / Clarification
		ging assumptions are present.				Status' statement (Section 2).
FNL2	None	No specific follow-up list was provided by the respondent despite FNL1 acknowledging assumptions exist.	AMBIGUOUS	Preserved as ambiguous stakeholder input; not converted into a specific requirement.	33. Open Issues / Clarifications	The consolidated Open Issues / Clarification register in Section 33 substitutes for this missing list, built from the gaps identified across all other fields.

39. Acceptance Criteria

Phase 1 may be considered ready for stakeholder acceptance review when, at minimum:

- Every CONFIRMED requirement in this document (Sections 9–20) is implemented and demonstrable against either the mock data contract (Track A, Section 24) or validated real MES data (Track B).
- Every capability whose underlying business definition is currently TBD (process loss, rework rate, WIP, first pass yield, quantity semantics — Section 14/20) is either validated and implemented, or explicitly deferred with stakeholder sign-off.
- SQL execution is verified, by testing, to be restricted to SELECT statements only via the dedicated read-only account, independent of any additional proposed safety layer.
- RBAC correctly restricts SQL/source visibility to Admin/IT users (UAT-08).
- Session-scoped conversational memory correctly resolves at least the follow-up pattern demonstrated in UAT-09.
- The profit question (UAT-05) is **not** answered with a fabricated value; the system instead communicates that the required financial data source is unconfirmed.

- The Open Issues register (Section 35) has been reviewed with IT/DBA/MES vendor/business stakeholders, and each item is either resolved or explicitly accepted as an open risk by the stakeholder-named approver (“Developer,” V3).
- Email and Microsoft Teams integration scope has been revisited with the stakeholder to either (a) obtain sufficient detail to implement a defined Phase 1 boundary, or (b) obtain explicit confirmation that they are deferred beyond Phase 1 — neither of which this SRS currently assumes on its own.

No acceptance criterion in this section asserts a specific numeric performance target, KPI formula, or timeline commitment beyond what the questionnaire itself supports.

40. Appendices

Appendix A — Classification Legend

Classification	Meaning
CONFIRMED	Explicitly stated by the stakeholder and sufficiently specific.
STAKEHOLDER-PROVIDED — REQUIRES TECHNICAL VERIFICATION	Provided by the stakeholder but requires validation by IT/DBA/MES developers.
AMBIGUOUS	An answer was given, but its meaning is unclear (e.g., “All,” “Over to you,” “IDK,” a range without units).
TBD	Information is missing and must be obtained before the related implementation decision can be finalized.
ASSUMPTION	A temporary assumption required to proceed, explicitly identified as such.
PROPOSED TECHNICAL SOLUTION	A design choice proposed by the implementation team/architect, not explicitly required by Renewsys.
FUTURE PHASE	Explicitly outside Phase 1 per the stakeholder’s own response.
OUT OF SCOPE	Explicitly excluded from the current implementation.

Appendix B — Corrections Made in This Revision (v3)

1. Email and Microsoft Teams are now kept as in-scope systems with TBD implementation detail (Section 22), corrected from any prior framing that moved them toward Future Phase.
2. The Source-of-Truth Register (Section 21) no longer uses “presumed”; Jinchen MES is described as a candidate, not-yet-validated source per domain.

3. Genuinely content-free questionnaire fields (H2/H3/H5, Q2, R3/R4, S2/S4, T4, U5/U6, V7/V8/V9) are marked **TBD — No response provided**, not “None”/“Not required”/“Out of Scope.”
4. The ambiguity between “MES server is in China, we do not handle” (B6) and “same company LAN / Direct LAN” (G1/G2) is now an explicit open technical issue (Section 12.4, Open Issue 3–5) rather than silently reconciled.
5. The traceability matrix (Section 38) is the complete 139-field matrix, replacing any earlier “representative sample” table, using the required column set: Field ID | Questionnaire Response | Interpretation | Classification | SRS Requirement/Treatment | SRS Section | Action/Clarification.
6. KPI formulas are restructured into a validation table (Section 14.2) with explicit “Confirmed by Renewsys?” = No for every row, and each formula labeled PROPOSED — REQUIRES BUSINESS VALIDATION.
7. FastAPI is explicitly reframed as “Proposed” technology, not a stakeholder mandate (Section 23.1), despite appearing in the questionnaire’s own question wording (G1).
8. Row-level security remains explicitly confirmed as **not** a Phase 1 requirement; F4 (“Fully accessible”) is clarified as not implying unrestricted, ungoverned querying (Section 12.2).
9. Performance figures (R2, R5) remain explicitly ambiguous rather than converted into precise numeric targets (Section 26).
10. The 24×7 vs. HA-not-required tension (R7, S1) is documented as an open issue requiring IT clarification, not resolved on Renewsys’s behalf (Section 27).

Appendix C — Statement of Completeness

All 139 questionnaire fields (A1–A9, B1–B10, C1–C6, D1–D13, E1–E8, F1–F6, G1–G9, H1–H5, I1–I5, J1–J4, K1–K5, L1–L6, M1–M4, N1–N4, O1–O4, P1–P3, Q1–Q3, R1–R7, S1–S5, T1–T4, U1–U6, V1–V9, FNL1–FNL2, plus Timestamp and Email Address) are represented in Section 38. None has been silently dropped, and no blank or ambiguous field has been converted into a negative or “out of scope” answer without explicit labeling as such.

End of Phase 1 Software Requirements Specification (Final Refined).